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A Unified Inference for Predictive Quantile Regression

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ABSTRACT

The asymptotic behavior of quantile regression inference becomes dramatically different when it involves a persistent predictor with zero or nonzero intercept. Distinguishing various properties of a predictor is empirically challenging. In this article, we develop a unified predictability test for quantile regression regardless of the presence of intercept and persistence of a predictor. The developed test is a novel combination of data splitting, weighted inference, and a random weighted bootstrap method. Monte Carlo simulations show that the new approach displays significantly better size and power performance than other competing methods in various scenarios, particularly when the predictive regressor contains a nonzero intercept. In an empirical application, we revisit the quantile predictability of the monthly S&P 500 returns between 1980 and 2019. Supplementary materials for this article are available online.

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1. Introduction

Predictive regression models are widely used in empirical economics and finance to investigate Granger causality, linear rational expectations, and market efficiency hypotheses. Due to the high degree of persistence existing in the regressor and the high contemporary correlation between the innovations of asset return and regressor, the standard *t*-test is invalid and inclined to over reject the null hypothesis of no predictability (see, e.g., Amihud and Hurvich 2004; Campbell and Yogo 2006). Because consistently estimating the persistent level is practically infeasible, it becomes empirically important to develop unified predictability tests regardless of the properties of predictive regressors. One attractive approach is the instrumental variable (dubbed as IVX) based inference, which was proposed by Magdalinos and Phillips (2009) and then extensively studied by Kostakis, Magdalinos, and Stamatogiannis (2015). This IVX-based Wald test is robust against various degrees of persistence of predictive regressors, ranging from purely stationary to purely nonstationary processes, in the presence of endogenous and heteroscedastic errors. Extensions of the IVX-based method are Phillips and Lee (2016), Yang et al. (2020), and Demetrescu and Rodrigues (2022). Another branch of the unified approach employs the weighted empirical likelihood method (Zhu, Cai, and Peng 2014; Liu et al. 2019; Yang et al. 2021), which always yields a Chi-squared limit regardless of the degree of persistence and embedded endogeneity in predictive regressors. Some recent development in the unified testing procedures includes Gungor and Luger (2020), Harvey, Lebourne, and Taylor (2021), Demetrescu and Rodrigues (2022), and Georgiev et al. (2021).

The literature above exclusively concentrates on the predictability of mean regression models. On the other hand, test-

ing the predictability of return for various quantile positions is also empirically meaningful. First, no predictability at the mean does not necessarily imply no predictability at all quantiles. For example, predictability might only exist for specific quantiles and is thus obscured by the no predictability at the mean. Also, the opposite sign of predictability for the lower and upper quantiles could lead to no predictability at the mean. Second, the predictability of return for different quantiles could be heterogeneous. For example, Gebka and Wohar (2013) report positive (negative) volume-return causality for high (low) return quantiles. Ma, Xiao, and Ma (2018) find investor's sentiment predicts aggregate stock returns at lower quantiles. Third, the predictability of the tail risk, such as Value-at-Risk, is crucial for risk managers and policymakers who are particularly interested in the predictability of return at tails. Fourth, predictive regressors which exhibit significant predicting power may vary among different quantiles (Cenesizoglu and Timmermann 2008; Gonzalo and Pitarakis 2012). These motivate researchers to investigate asset return predictability for different quantiles via a predictive quantile regression.

Quantile regression (QR), first introduced by Koenker and Bassett (1978), has become a standard tool to model the conditional quantiles of a dependent variable as a function of regressors. By regressing the return on predictive regressors at a sequence of quantile points, predictive QR provides complete information about the predictability of the return distribution rather than at the mean. Prior studies try to investigate the predictability of conditional quantiles via QR when the degrees of persistence of regressors are known. For example, Xiao (2009) studies the inference of quantile cointegrating regression in the presence of unit root regressors. Maynard, Shimotsu, and Wang (2011) use Bonferroni's method to conduct inference in

predictive QR when the predictive regressor has a near unit root. Recently, Xu (2021) shows that the inference of multi-horizon QR is more robust than the standard multi-horizon mean regression when the predictive regressor is stationary.

Like unified tests for predictive mean regression with unknown persistence in predictive regressors, recent researchers seek unified tests in predictive QR regardless of the persistence of the predictive regressor. A seminal work by Lee (2016) extends the IVX-based statistic to QR and proposes the IVX-QR method for a unified inference. He argues that the proposed IVX-QR method is uniformly valid over various persistent levels, ranging from stationarity to mild explosiveness, in the presence of multiple persistent predictive regressors. Simulation results show IVX-QR corrects the size distortion from persistent multivariate predictive regressors while preserving discriminatory power. To accommodate the conditional heteroscedastic innovations in the predictive QR model, Fan and Lee (2019) adopt the moving block bootstrap (MBB) technology and propose an MBB based IVX-QR approach (MBB-QR hereafter) to assure valid inference at both tails. Simulation results show that this MBB-based method not only maintains the good properties of IVX-QR but also reduces the inferential errors by avoiding nuisance parameter estimation when conditional heteroscedasticity exists in error terms. In a recent study, Cai, Chen, and Liao (2022) develop an inferential theory based on a double-weighted method for predictive QR (DW-QR hereafter) and show the robustness of their method when the predictive regressors are either stationary or nearly integrated.

A common assumption of the three aforementioned unified predictive QR inferences is that predictors follow AR models with zero intercepts. However, when examining the eleven financial and macroeconomic variables in the empirical example, we find evidence that most predictors contain significant nonzero intercepts at the conventional levels. Moreover, as demonstrated in the simulation, IVX-QR and IVX-MBB suffer severe size distortion when the predictive regressor has a nonzero intercept. At the same time, DW-QR becomes slightly oversized at tails when a nonzero intercept is present. Finally, from the theoretical perspective, it is known that a unit root AR(1) sequence diverges with a faster divergence rate in the case of a nonzero intercept than in the case of a zero intercept. Hence, the instrumental variables constructed by Lee (2016), Fan and Lee (2019), and Cai, Chen, and Liao (2022) for zero intercepts in modeling predictors behave differently from the case of nonzero intercepts. Consequently, it remains unknown if these three unified inferences can unify the cases of zero and nonzero intercepts in predictors.

In this study, we develop a novel unified test in the predictive QR model regardless of the presence of intercept and persistence of a predictor. Specifically, we employ a technique of weighted estimation in the predictive QR model to handle the effect of the nonstationary predictive regressor, as in Chan, Li, and Peng (2012). Because the weighted QR equations would behave asymptotically equivalently if using the full observations for the case of nonzero intercepts, we propose to use the idea of data splitting to overcome this difficulty, which always results in a normal limit after a random normalization. Since the random normalization and the variance in the normal limit depend on different cases, a direct inference is infeasible. Therefore,

we employ a random weighted bootstrap method to estimate them in a unified way, leading to a unified predictability test via the square of a t -test statistic. We call this new method a random weighted bootstrap predictive QR (RWB-QR) method and show that its test statistic achieves a Chi-squared limit under the null hypothesis of no predictability regardless of intercept and persistence in the predictive regressor. As the test neither infers the heteroscedastic errors nor generates samples from the fitted model via resampling residuals like the residual-based bootstrap methods, the proposed RWB-QR test is robust against heteroscedastic errors in the predictive QR model and computationally convenient.¹

We conduct extensive Monte Carlo simulations to examine the finite sample performance of the developed RWB-QR and find that our method yields robust inferential results under various scenarios. In particular, RWB-QR exhibits excellent size control, especially when the predictive regressor contains a nonzero intercept. At the same time, IVX-QR and IVX-MBB have severe size distortion, and DW-QR becomes slightly oversized at tails. The power performance of our method is also uniformly better than the three competing methods under different scenarios when the predictive regressor is highly persistent. In an empirical example, we revisit the monthly stock market return predictability in the United States between 1980 and 2019 with the commonly used financial and macroeconomic predictors in the literature. The empirical result is broadly consistent with our findings in the simulation part. It shows that most predictors do not exhibit predicting power on the stock return for all quantile positions by RWB-QR, except for the earnings-price ratio and net equity expansion, which are significant for the lower quantiles. This result is notably different from IVX-QR, which indicates that many predictors display significant predicting power. The lower and upper tail quantiles' predictability is substantially stronger than the middle quantiles. MBB-QR and DW-QR also yield more significant predictors than RWB-QR at the tails. Further analyses indicate that such a different result between RWB-QR and the three competing methods can be partly explained by the presence of nonzero intercepts in most of these predictors.

The remaining structure of the article is as follows. [Section 2](#) introduces the proposed method and provides asymptotic results. [Section 3](#) reports the finite-sample simulation comparison results. [Section 4](#) applies the proposed RWB-QR method to investigate the predictive ability of 11 economic and financial variables on the monthly S&P 500 return between 1980 and 2019. We conclude in [Section 5](#). The technical proofs and additional simulation results are provided in the supplementary appendix.

2. Models, Methodologies, and Theoretical Results

Let y_t denote a predicted variable such as stock return, and x_t be a predictive variable such as dividend-price ratio or treasury bill rate. We consider the linear predictive QR model as

¹It is worth mentioning that Zhu, Cai, and Peng (2014) also develop a data splitting strategy, which can handle the intercept effect using the technique of differencing over long-range observations for an ordinary predictive regression model. However, their method can not be applied to heteroscedastic errors.

$$\begin{cases} y_t = \alpha_\tau + \beta_\tau x_{t-1} + \varepsilon_{t\tau}, & \varepsilon_{t\tau} = u_t - \sigma_t Q_\eta(\tau), \\ u_t = \sigma_t \eta_t, & \sigma_t^2 = w + \sum_{i=1}^q a_i u_{t-i}^2 + \sum_{j=1}^r b_j \sigma_{t-j}^2, \end{cases} \quad (1)$$

for $t = 1, \dots, T$, where $P(\eta_t \leq Q_\eta(\tau)) = \tau$ indicates $P(\varepsilon_{t\tau} \leq 0 | \mathcal{F}_{t-1}) = \tau$ with $\mathcal{F}_t$ being the sigma field generated by $\{(\eta_k, \eta_{x,k})\}_{k \leq t}$, that is, $\alpha_\tau + \beta_\tau x_{t-1}$ is the τ th conditional quantile of y_t given x_{t-1} . The predictive variable follows

$$\begin{cases} x_t = \mu + \phi x_{t-1} + e_t, \\ e_t = \sum_{i=0}^{\infty} \theta_i v_{t-i}, & v_t = \sigma_{x,t} \eta_{x,t}, \quad \sigma_{x,t}^2 = w_x \\ + \sum_{i=1}^{q_x} a_{x,i} v_{t-i}^2 + \sum_{j=1}^{r_x} b_{x,j} \sigma_{x,t-j}^2, \end{cases} \quad (2)$$

where $\{(\eta_t, \eta_{x,t})\}_{t=1}^T$ is a sequence of independent and identically distributed random vectors with means zero and variances one, and $\alpha_\tau, \beta_\tau, w, a_i, b_j, \mu, \phi, \theta_i, w_x, a_{x,i}$ and $b_{x,j}$ are unknown parameters. Because our developed method below does not infer the GARCH models, it could be naturally extended to other heteroscedastic error scenarios as those in Zhu and Ling (2015).

The QR estimation for α_τ and β_τ minimizes

$$\sum_{t=1}^T \rho_\tau(y_t - \alpha_\tau - \beta_\tau x_{t-1}), \quad (3)$$

where $\rho_\tau(u) = u(\tau - I(u < 0))$. It is known that the limit of the QR estimation is normal for stationary $\{x_t\}$ and nonnormal for persistent $\{x_t\}$; see Theorem 3.1 of Fan and Lee (2019). To unify these two cases, Lee (2016) and Fan and Lee (2019) replace x_{t-1} in (3) with a less persistent instrumental variable, which sacrifices the rate of convergence in the persistent case by comparing Theorems 3.1 and 3.2 in Fan and Lee (2019). Alternatively, Cai, Chen, and Liao (2022) propose to replace $\beta_\tau x_{t-1}$ in (3) by $\beta_\tau(x_{t-1} - z_{t-1}) + \gamma_\tau z_{t-1}$, where the auxiliary variable z_{t-1} is a combination of Type-II instrumental variable (simulated random walks) and $x_{t-1}/\sqrt{1 + x_{t-1}^2}$. All these three predictability tests assume $\mu = 0$ in (2) and remain unknown if they unify the cases of zero and nonzero intercept in modeling predictors.

To develop a unified predictability test for $H_0 : \beta_\tau = 0$ regardless of the properties of $\{x_t\}$, we split the data into two parts and use one part to formulate the score function for α_τ and the other part to construct a weighted score function for β_τ . In this way, the weighted estimator for β_τ after a random normalization has a normal limit regardless of the properties of $\{x_t\}$. Still, the asymptotic variance and the random normalization depend on the properties of $\{x_t\}$, and the random normalization does not have a constant sign. Hence, to construct a valid test, we use the square of the estimator for β_τ to get rid of the sign issue and adopt a random weighted bootstrap method to estimate its asymptotic variance in a unified way. More specifically, let $m = \lfloor T/2 \rfloor$ with $\lfloor \cdot \rfloor$ being the integer operator and define

$$w_t = \frac{x_{t-1}}{\sqrt{1 + x_{t-1}^2}}. \quad (4)$$

We estimate $\hat{\alpha}_\tau$ and $\hat{\beta}_\tau$ by solving the following two score equations:

$$\begin{cases} \sum_{t=1}^m \psi_\tau(y_t - \alpha_\tau - \beta_\tau x_{t-1}) = 0, \\ \sum_{t=m+1}^{2m} \psi_\tau(y_t - \alpha_\tau - \beta_\tau x_{t-1}) w_t = 0, \end{cases} \quad (5)$$

where $\psi_\tau(u) = \tau - I(u < 0)$. Note that if we do not split the sample, the above two equations will be asymptotically equivalent when $\{x_t\}$ is nonstationary with nonzero intercepts, leading to a degenerate limit of the weighted estimation.

We employ the following regularity conditions to derive the asymptotic limit of the above estimators.

C1. The predictive variable x_t , with the initial value of x_0 being constant, belongs to one of the following persistence classes:

Case (i): $|\phi| < 1$. That is, $\{x_t\}$ is a stationary AR(1) process.

Case (ii): $\mu = 0$ and $\phi = 1 - c/T$ for some constant c . That is, $\{x_t\}$ is a nearly integrated AR(1) process with zero intercepts for some $c \neq 0$. When $c = 0$, $\{x_t\}$ is a unit root AR(1) process.

Case (iii): $\mu \neq 0$ and $\phi = 1 - c/T$ for some constant c . That is, $\{x_t\}$ is a nearly integrated AR(1) process with a nonzero intercept.

C2. $\{\sigma_t\}$ and $\{\sigma_{x,t}\}$ are strictly stationary with finite variance, w and w_x are positive, and $a_i, b_j, a_{x,i}$ and $b_{x,j}$ are nonnegative.

C3. $E(|\eta_t|^{2+c_0} + |\eta_{x,t}|^{2+c_0}) < \infty$ for some constant $c_0 > 0$.

C4. $\{e_t\}$ is strictly stationary with $\sum_{j=0}^{\infty} \theta_j^2 < \infty$, that is, finite variance, and $\theta_0 = 1$.

C5. The sequence of stationary probability density functions $\{f_{\varepsilon,t-1}(s)\}$ of ε_t conditional on $\mathcal{F}_{t-1}$ satisfies a moment condition with a nondegenerate mean $f_\varepsilon(0) = E(f_{\varepsilon,t-1}(0)) > 0$ and $E(f_{\varepsilon,t-1}^{d_0}(0)) < \infty$ for some $d_0 > 1$.

C6. For each t and $\tau \in (0, 1)$, $|f'_{\varepsilon,t-1}(s)| < \infty$ and $f_{\varepsilon,t-1}(s) < \infty$ almost surely for all $|s| < d_0^*$ for some $d_0^* > 0$.

These conditions are quite mild, and C4–C6 are also assumed in Cai, Chen, and Liao (2022). Note that $\{\sigma_t\}$ is strictly stationary, $f_\varepsilon(0)$ is independent of t . For convenience, hereafter we write $S_{[2ms]}^x = \frac{1}{\sqrt{m}} \sum_{t=1}^{[2ms]} e_t$ and $S_{[2ms]}^y = \frac{1}{\sqrt{m}} \sum_{t=1}^{[2ms]} \psi_\tau(\varepsilon_t)$ for some $s \in (0, 1]$. Then, for Case (ii), it follows from Cai, Chen, and Liao (2022) that

$$\begin{pmatrix} S_{[2ms]}^x \\ S_{[2ms]}^y \end{pmatrix} \xrightarrow{D} W(s) := \begin{pmatrix} W_x(s) \\ W_y(s) \end{pmatrix}, \quad (6)$$

$$\frac{1}{\sqrt{m}} x_{[2ms]} \xrightarrow{D} J_c^x(s) = \int_0^s e^{(s-s_1)c} dW_x(s_1) \quad \text{in the space } D[0, 1], \quad (7)$$

where $(W_x(s), W_y(s))^\top$ is a bivariate Gaussian processes with covariance matrix

$$2 \times \begin{pmatrix} \sum_{h=-\infty}^{\infty} E(v_t v_{t+h}) & \sum_{h=-\infty}^{\infty} E(\psi_\tau(\varepsilon_t) v_{t+h}) \\ \sum_{h=-\infty}^{\infty} E(\psi_\tau(\varepsilon_t) v_{t+h}) & \tau(1-\tau) \end{pmatrix}$$

and $D[0, 1]$ is the space of real-valued functions on the interval $[0, 1]$ that are right continuous with finite left limits. We refer to Phillips (1987) and Cai, Chen, and Liao (2022) for details. In the sequel, we denote “ $\xrightarrow{D}$ ”, “ $\xrightarrow{d}$ ”, “ $\xrightarrow{P}$ ” as the convergence in space, in distribution, and in probability, respectively. Also, $\alpha_{\tau,0}$ and $\beta_{\tau,0}$ denote the true values of α_τ and β_τ , respectively.

For the weighted estimation above, we have the following theorem.

Theorem 1. Suppose models (1) and (2) satisfy conditions C1–C6.

For Case (i), as $T \rightarrow \infty$,

$$D_{1,m} \left(\frac{\sqrt{m}(\hat{\alpha}_\tau - \alpha_{\tau,0})}{\sqrt{m}(\hat{\beta}_\tau - \beta_{\tau,0})} \right) \xrightarrow{d} N(0, \Sigma_1)$$

and $A_{1,m} \times \sqrt{m}(\hat{\beta}_\tau - \beta_{\tau,0}) \xrightarrow{d} N(0, \gamma_1^\top \Sigma_1 \gamma_1)$, where

$$D_{1,m} = f_\varepsilon(0) \begin{pmatrix} 1 & \frac{1}{m} \sum_{t=1}^m x_{t-1} \\ \frac{1}{m} \sum_{t=m+1}^{2m} w_t & \frac{1}{m} \sum_{t=m+1}^{2m} w_t x_{t-1} \end{pmatrix},$$

$$\Sigma_1 = \tau(1-\tau) \begin{pmatrix} 0 \\ 0 \ E \left(\frac{x_1^2}{1+x_1^2} \right) \end{pmatrix},$$

$$\gamma_1 = \left(-E \left(\frac{x_1}{\sqrt{1+x_1^2}} \right), 1 \right),$$

$$A_{1,m} = \left\{ \frac{1}{m} \sum_{t=m+1}^{2m} x_{t-1} w_t - \frac{1}{m} \sum_{t=m+1}^{2m} w_t \frac{1}{m} \sum_{t=1}^m x_{t-1} \right\} \times f_\varepsilon(0).$$

For Case (ii), as $T \rightarrow \infty$,

$$D_{2,m} \left(\frac{\sqrt{m}(\hat{\alpha}_\tau - \alpha_{\tau,0})}{m(\hat{\beta}_\tau - \beta_{\tau,0})} \right) \xrightarrow{d} \left(\text{sgn}(J_c^x(1)) W_y(1) - \text{sgn} \left(J_c^x \left(\frac{1}{2} \right) \right) W_y \left(\frac{1}{2} \right) \right)$$

and

$$A_{2,m} \times m(\hat{\beta}_\tau - \beta_{\tau,0}) \xrightarrow{d} \text{sgn}(J_c^x(1)) \left\{ W_y(1) - 2W_y \left(\frac{1}{2} \right) \right\},$$

where $\text{sgn}(\cdot)$ denotes the sign function,

$$D_{2,m} = f_\varepsilon(0) \begin{pmatrix} 1 & \frac{1}{m} \sum_{t=1}^m \frac{x_{t-1}}{\sqrt{m}} \\ \frac{1}{m} \sum_{t=m+1}^{2m} w_t & \frac{1}{m} \sum_{t=m+1}^{2m} w_t \frac{x_{t-1}}{\sqrt{m}} \end{pmatrix},$$

$$A_{2,m} = \left\{ \frac{1}{m} \sum_{t=m+1}^{2m} \frac{x_{t-1}}{\sqrt{m}} w_t - \frac{1}{m} \sum_{t=m+1}^{2m} w_t \frac{1}{m} \sum_{t=1}^m \frac{x_{t-1}}{\sqrt{m}} \right\} \times f_\varepsilon(0).$$

For Case (iii), as $T \rightarrow \infty$,

$$D_{3,m} \left(\frac{\sqrt{m}(\hat{\alpha}_\tau - \alpha_{\tau,0})}{m^{3/2}(\hat{\beta}_\tau - \beta_{\tau,0})} \right) \xrightarrow{d} N(0, \Sigma_3)$$

and $A_{3,m} \times m^{3/2}(\hat{\beta}_\tau - \beta_{\tau,0}) \xrightarrow{d} N(0, 2\tau(1-\tau))$, where

$$D_{3,m} = f_\varepsilon(0) \begin{pmatrix} 1 & \frac{1}{m} \sum_{t=1}^m \frac{x_{t-1}}{m} \\ \frac{1}{m} \sum_{t=m+1}^{2m} w_t & \frac{1}{m} \sum_{t=m+1}^{2m} w_t \frac{x_{t-1}}{m} \end{pmatrix},$$

$$\Sigma_3 = \begin{pmatrix} \tau(1-\tau) & 0 \\ 0 & \tau(1-\tau) \end{pmatrix},$$

$$A_{3,m} = \left\{ \frac{1}{m} \sum_{t=m+1}^{2m} \frac{x_{t-1}}{m} w_t - \frac{1}{m} \sum_{t=m+1}^{2m} w_t \frac{1}{m} \sum_{t=1}^m \frac{x_{t-1}}{m} \right\} \times f_\varepsilon(0).$$

Theorem 1 above shows that $\hat{\beta}_\tau - \beta_{\tau,0}$ after a random normalization with a sign function for Case (ii) can be approximated by a normal distribution with mean zero. Because the random normalization and the variance in the normal limit depend on the properties of $\{x_t\}$, we can not directly employ **Theorem 1** to test predictability in a unified way. Therefore, we propose to use the random weighted bootstrap method (Jin, Ying, and Wei 2001; Zhu 2016) to estimate the asymptotic variance of $\hat{\beta}_\tau$ in a unified way without any prior knowledge about the properties of $\{x_t\}$. Since the random weighted bootstrap method neither infers the heteroscedastic errors nor generates samples from the fitted model via resampling residuals like residual-based bootstrap methods, the developed test is robust against conditional heteroscedasticity and computationally convenient.

Step (A1): Draw a random sample with sample size T from a distribution function with mean one and variance one, say the standard exponential distribution. Denote the generated random sample by $\delta_1^b, \dots, \delta_T^b$.

Step (A2): Solve the following two equations to get estimators $\hat{\alpha}_\tau^b$ and $\hat{\beta}_\tau^b$ for α_τ and β_τ :

$$\begin{cases} \sum_{t=1}^m \delta_t^b \psi_\tau(y_t - \alpha_\tau - \beta_\tau x_{t-1}) = 0, \\ \sum_{t=m+1}^{2m} \delta_t^b \psi_\tau(y_t - \alpha_\tau - \beta_\tau x_{t-1}) w_t = 0. \end{cases} \quad (8)$$

Step (A3): Repeat the above two steps B times to get $\left\{ (\hat{\alpha}_\tau^b, \hat{\beta}_\tau^b)^\top \right\}_{b=1}^B$.

Since the random weighted bootstrap method does not generate samples from the fitted model via resampling residuals, its computational burden is relatively minimal, allowing a larger value of B . We employ $B = 5000$ in the simulation study below, giving robust size and power.

The theorem below derives the limit of $(\hat{\alpha}_\tau^b - \hat{\alpha}_\tau, \hat{\beta}_\tau^b - \hat{\beta}_\tau)^\top$, showing that $|\hat{\beta}_\tau^b - \hat{\beta}_\tau|$ and $|\hat{\beta}_\tau - \beta_{\tau,0}|$ have the same limit, which leads to a unified predictability test.

Theorem 2. Suppose models (1) and (2) satisfy conditions C1–C6.

For Case (i), as $T \rightarrow \infty$,

$$D_{1,m} \left(\frac{\sqrt{m}(\hat{\alpha}_\tau^b - \hat{\alpha}_\tau)}{\sqrt{m}(\hat{\beta}_\tau^b - \hat{\beta}_\tau)} \right) \xrightarrow{d} N(0, \Sigma_1)$$

and $A_{1,m} \times \sqrt{m}(\hat{\beta}_\tau^b - \hat{\beta}_\tau) \xrightarrow{d} N(0, \gamma_1^\top \Sigma_1 \gamma_1)$.

For Case (ii), as $T \rightarrow \infty$,

$$D_{2,m} \left(\frac{\sqrt{m}(\hat{\alpha}_\tau^b - \hat{\alpha}_\tau)}{m(\hat{\beta}_\tau^b - \hat{\beta}_\tau)} \right) \xrightarrow{d} \left(\text{sgn}(J_c^x(1)) W_y^*(1) - \text{sgn} \left(J_c^x \left(\frac{1}{2} \right) \right) W_y^* \left(\frac{1}{2} \right) \right)$$

and

$$A_{2,m} \times m(\hat{\beta}_\tau^b - \hat{\beta}_\tau) \xrightarrow{d} \text{sgn}(J_c^x(1)) \left\{ W_y^*(1) - 2W_y^* \left(\frac{1}{2} \right) \right\},$$

where $W_y^*(s)$ is a Gaussian process having the same covariance structure as that of $W_y(s)$ but independent of both $W_x(s)$ and $W_y(s)$.

For Case (iii), as $T \rightarrow \infty$,

$$\mathbf{D}_{3,m} \left(\frac{\sqrt{m}(\hat{\alpha}_\tau^b - \hat{\alpha}_\tau)}{m^{3/2}(\hat{\beta}_\tau^b - \hat{\beta}_\tau)} \right) \xrightarrow{d} N(0, \Sigma_3)$$

$$\text{and } A_{3,m} \times m^{3/2}(\hat{\beta}_\tau^b - \hat{\beta}_\tau) \xrightarrow{d} N(0, 2\tau(1 - \tau)).$$

Theorem 3. Under conditions of [Theorem 1](#), we have

$$\frac{(\hat{\beta}_\tau - \beta_{\tau,0})^2}{\frac{1}{B} \sum_{b=1}^B (\hat{\beta}_\tau^b - \hat{\beta}_\tau)^2} \xrightarrow{d} \chi_1^2$$

as $T \rightarrow \infty$ and $B \rightarrow \infty$, where χ_1^2 is a Chi-squared distribution with one degree of freedom.

Using the above theorem, our proposed random weighted bootstrap based Wald statistic for testing $H_0 : \beta_\tau = 0$ is

$$\hat{W} = \frac{\hat{\beta}_\tau^2}{\frac{1}{B} \sum_{b=1}^B (\hat{\beta}_\tau^b - \hat{\beta}_\tau)^2}, \quad (9)$$

and we reject the null hypothesis at level $a \in (0, 1)$ whenever $\hat{W} > \chi_{1,1-a}^2$, where $\chi_{1,1-a}^2$ is the $(1 - a)$ th quantile of the Chi-squared distribution with one degree of freedom.

Remark 1. As can be seen from the proofs of [Theorems 1–3](#) in the supplementary appendix, the purpose of adding $1/\sqrt{1 + x_{t-1}^2}$ into w_t is to control the order of x_{t-1} and ensure that $w_m \xrightarrow{p} \text{sgn}(J_c^x(1/2))$, which is random, but $|w_m| \xrightarrow{p} 1$ in Case (ii). Because the two equations in (5) for a persistent predictor would behave asymptotically equivalently without data splitting in Case (iii), it is necessary to use this data-splitting strategy to ensure a nondegenerate limit. This data-splitting method may lose efficiency but is still more efficient than the IVX-based QR method, which has a slower convergence rate due to the employed less persistent instrumental variable.

Remark 2. When the predictive QR has multiple predictors, possible nonstationary $\{x_t\}$ and stationary $\{z_{t,i}\}$ for $i = 1, \dots, p$, in the following model:

$$y_t = \alpha_\tau + \beta_\tau x_{t-1} + \sum_{i=1}^p \phi_{\tau,i} z_{t-1,i} + \varepsilon_t,$$

where ε_t takes the same form as in (1), one may use the following weighted score equations,

$$\begin{cases} \sum_{t=1}^m \psi_\tau(y_t - \alpha_\tau - \beta_\tau x_{t-1} - \sum_{i=1}^p \phi_{\tau,i} z_{t-1,i}) = 0, \\ \sum_{t=1}^m \psi_\tau(y_t - \alpha_\tau - \beta_\tau x_{t-1} - \sum_{i=1}^p \phi_{\tau,i} z_{t-1,i}) z_{t-1,i} = 0, \\ i = 1, 2, \dots, p, \\ \sum_{t=m+1}^{2m} \psi_\tau(y_t - \alpha_\tau - \beta_\tau x_{t-1} - \sum_{i=1}^p \phi_{\tau,i} z_{t-1,i}) w_t = 0, \end{cases}$$

to obtain a similar predictability test for the null hypothesis $(\beta_\tau, \phi_{\tau,1}, \phi_{\tau,2}, \dots, \phi_{\tau,p})^\top = \mathbf{0}$. This test will have a Chi-squared limit with $p + 1$ degrees of freedom under the null hypothesis of no predictability. A simulation study shows that this test has an accurate size and good powers.

3. Simulation

In this section, we examine the finite sample size and power performance of the proposed RWB-QR method. Specifically, we generate data from

$$y_t = \alpha_\tau + \beta_\tau x_{t-1} + u_{t\tau}, \quad x_t = \mu + (1 - c/T)x_{t-1} + u_{xt},$$

where the intercept α_τ equals 0.08 for all τ , and μ equals either 0 (zero intercepts) or 0.05 (nonzero intercept), which are empirically relevant, as will be shown in the next section.² To adjust the degree of persistence in predictor x_t , we allow $c \in \{0, 2, 5, 10, 100\}$, where $c = 0$ indicates the exact unit root process. Increasing the value of $|c|$ will lower the degree of persistence of the predictor x_t . The sample size $T \in \{500, 1000\}$. As for the structure of the two error terms u_t and u_{xt} , similar to Fan and Lee (2019), we consider the following two scenarios.

(i). IID CASE

$$\begin{pmatrix} u_t \\ u_{xt} \end{pmatrix} \sim \text{iid } N(\mathbf{0}, \Sigma), \text{ where } \mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ and } \Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

(ii). CONDITIONAL HETEROSCEDASTICITY CASE

$$u_t \text{ follows a GARCH}(1,1) \text{ process such that } u_t = \sigma_t \eta_t \text{ and } \sigma_t^2 = \omega_0 + \omega_1 u_{t-1}^2 + \omega_2 \sigma_{t-1}^2, \text{ and } \begin{pmatrix} \eta_t \\ u_{xt} \end{pmatrix} \sim \text{iid } N(\mathbf{0}, \Sigma)$$

with $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. To render empirical relevance, we assume $\omega_0 = 0.1$, $\omega_1 = 0.1$, and $\omega_2 = 0.57$ throughout the simulations.

In both scenarios, ρ denotes the correlation coefficient that determines the degree of endogeneity, and, similar to Fan and Lee (2019), we set $\rho = -0.95$ so that u_t and u_{xt} are highly correlated. Furthermore, we let $u_{t\tau} \equiv u_t - F^{-1}(\tau)$ where $F(\cdot)$ denotes the cumulative distribution function of $N(0, \sigma_t^2)$. We repeat each simulation 1000 times with $B = 5000$ bootstrapped samplings in each simulation and then compare the size and power performance of the proposed RWB-QR with that of IVX-QR by Lee (2016), MBB-QR by Fan and Lee (2019), and DW-QR by Cai, Chen, and Liao (2022).³

We first compare the size performance of the four methods with varying degrees of persistent predictors for quantile levels $\tau \in \{0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95\}$. That is, $\beta_\tau = 0$. [Figure 1](#) presents the size performance under the IID setting when the predictor does not contain an intercept (i.e., $\mu = 0$), and the displayed values denote the estimated rejection rates over 1000 simulations. It shows that RWB-QR exhibits excellent size control across all quantiles and degrees of persistence when $T = 500$ (upper panel) and $T = 1000$ (lower panel). On the contrary, IVX-QR is severely oversized at the tails ($\tau = 0.05, 0.1, 0.9, 0.95$) when the predictor is highly persistent ($c = 0$). However, increasing the sample size from 500 to 1000 slightly improves the size control by alleviating the sparsity at the

²We also examine the scenario, which allows α_τ to vary at different quantile levels and find similar results. The detailed results are available upon request. We thank one anonymous reviewer for suggesting this.

³To apply IVX-QR, we generate the instrument $z_t = R_{nz} z_{t-1} + \Delta x_t$ where $\Delta x_t = x_t - x_{t-1}$, $R_{nz} = I_k + n^\delta C_z$ and $C_z = c_z I_k$, with I_k being a $k \times k$ identity matrix, $c_z = -5$ as suggested by Lee (2016), and δ selected by the practical rule in Lee (2016). To apply MBB-QR, we generate 1000 random samples in each simulation and set block length $b = \lceil n^{1/4} \rceil$ where $\lceil a \rceil$ denotes the least integer that is greater or equal to a .

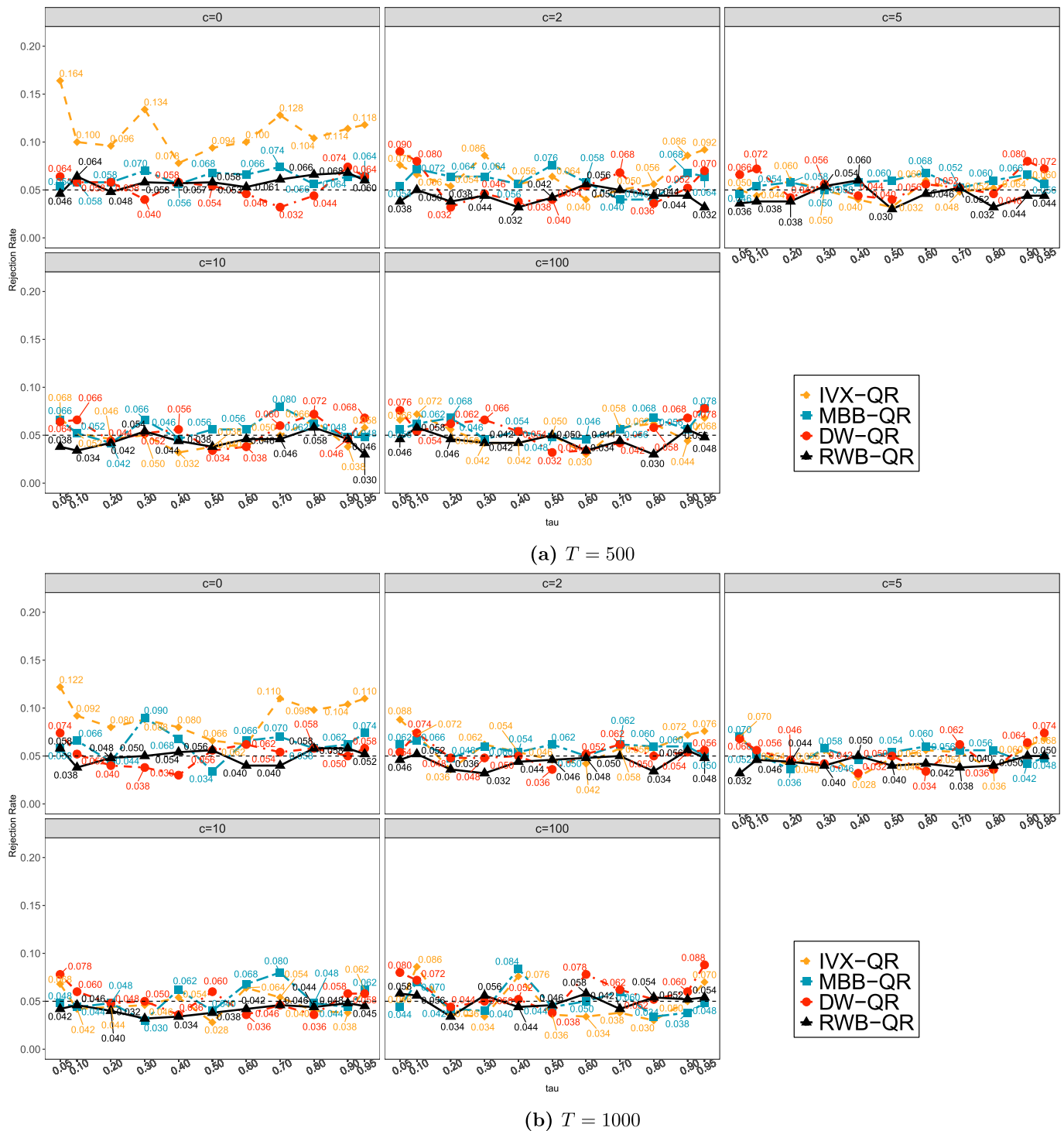


Figure 1. Comparison of size: IID and zero intercept ($\mu = 0$).

NOTE: This figure displays the size performance of RWB (black), IVX (yellow), MBB (blue), and DW (red) for different τ -quantiles under the IID setting with a zero intercept ($\mu = 0$) in the predictor. The sample size $T = 500$ (upper panel) and 1000 (lower panel). For RWB, the number of bootstraps is 5000. Each simulation is repeated 1000 times.

tails when the sample size is small. This finding is in line with Lee (2016), who also documents the size distortion of IVX-QR at the tails. MBB-QR and DW-QR display reasonably good size performance across all quantiles and degrees of persistence. When a nonzero intercept (i.e., $\mu = 0.05$) is added, Figure 2 compares the size performance of the four methods under IID. RWB-QR still shows excellent size control throughout all quantile positions regardless of the degree of persistence of the predictor

and the sample size. At the same time, the oversizing exhibited by IVX-QR becomes even more severe. In particular, the largest rejection rate produced by IVX-QR is as large as 0.2 when $c = 0$ and $\tau = 0.3$. MBB-QR also exhibits slight oversizing at the lower tail when the predictor is a unit root process, and the size distortion becomes more severe when $T = 1000$. DW-QR still displays good size control overall but is slightly oversized at the tails when $c = 0$.

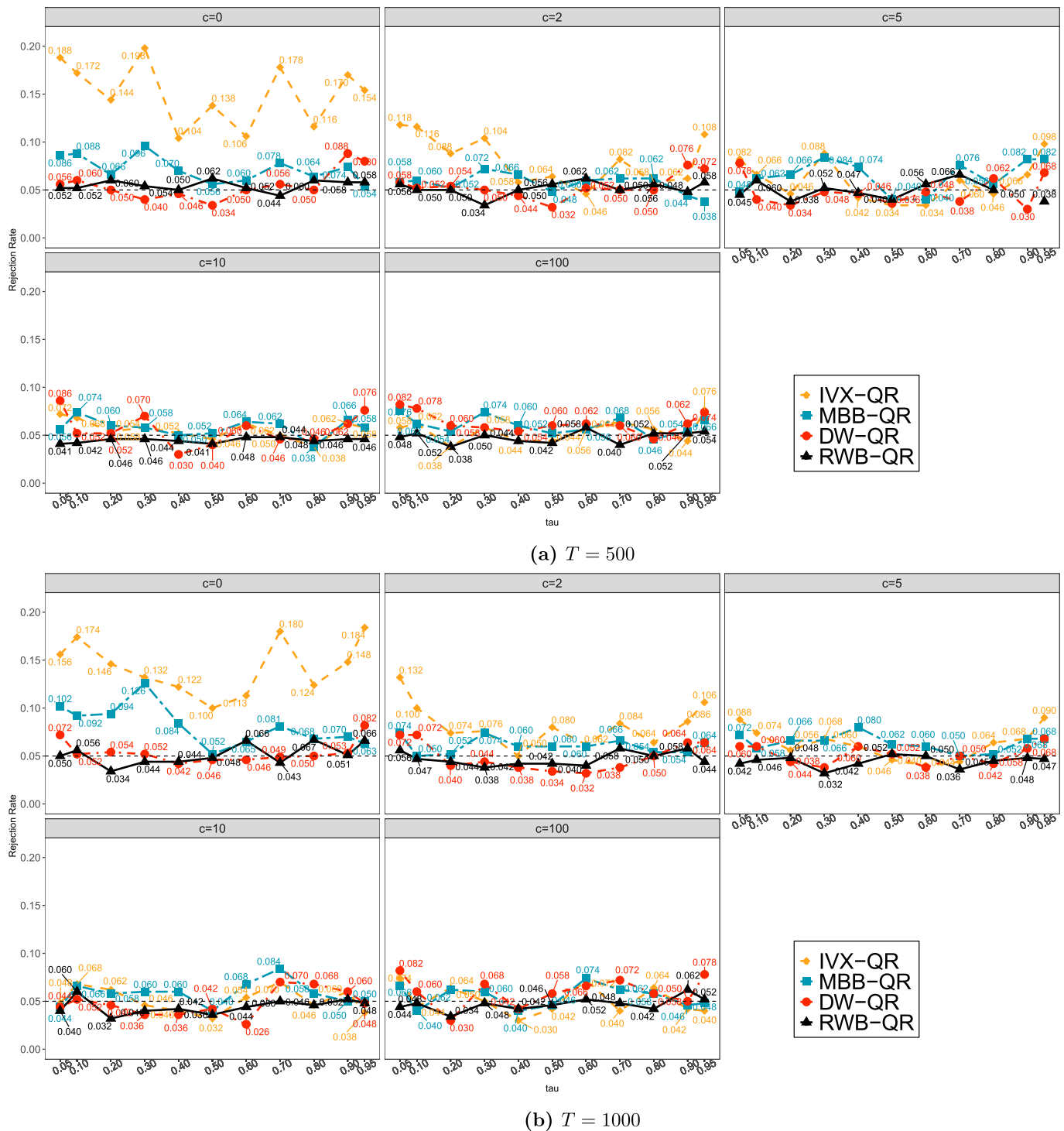


Figure 2. Comparison of size: IID and nonzero intercept ($\mu = 0.05$).

NOTE: This figure displays the size performance of RWB (black), IVX (yellow), MBB (blue), and DW (red) for different τ -quantiles under the IID setting with a nonzero intercept ($\mu = 0.05$) in the predictor. The sample size $T = 500$ (upper panel) and 1000 (lower panel). For RWB, the number of bootstraps is 5000. Each simulation is repeated 1000 times.

Next, we compare the four methods' size performance under the conditional heteroscedasticity setting. In Figure 3, which displays the zero intercept case, we can observe excellent size control by RWB-QR and DW-QR across all quantiles and degrees of persistence of the predictor. At the same time, IVX-QR exhibits severe oversizing when the predictor is a pure unit root process ($c = 0$). The size performance by MBB-QR is also reasonably

good, but it is slightly oversized when the predictor is highly persistent. However, introducing a nonzero intercept notably inflates the size distortion by IVX-QR and MBB-QR across all quantiles when the predictor is highly persistent. At the same time, RWB-QR and DW-QR still present excellent size controls, as displayed by both panels in Figure 4. Such deterioration is apparent when $c = 0$. For IVX-QR, the largest empirical size is

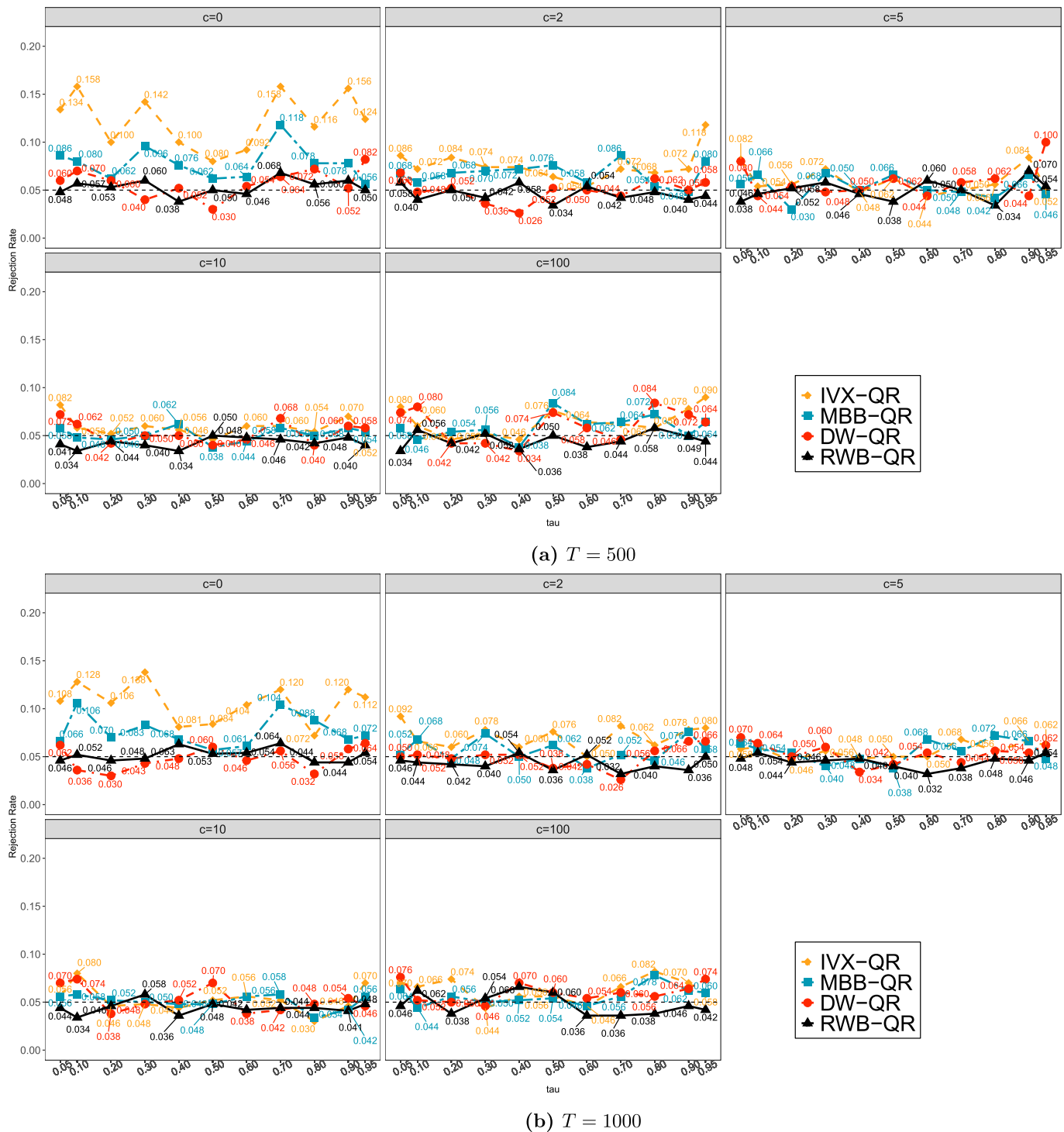


Figure 3. Comparison of size: Conditional heteroscedasticity and zero intercept ($\mu = 0$).

NOTE: This figure displays the size performance of RWB (black), IVX (yellow), MBB (blue), and DW (red) for different τ -quantiles under the conditional heteroscedasticity setting with a zero intercept ($\mu = 0$) in the predictor. The sample size $T = 500$ (upper panel) and 1000 (lower panel). For RWB, the number of bootstraps is 5000. Each simulation is repeated 1000 times.

as large as 0.206 when $\tau = 0.1$, a 54% increase in rejection rate compared with its counterpart when there is no intercept. Even when $\tau = 0.5$, at which IVX-QR exhibits relatively good size performance when $c = 0$, the inclusion of the nonzero intercept makes the size distortion remarkably larger. For MBB-QR, we also observe severe oversizing when $c = 0$ in both panels.

In the final step, we investigate the local power of the RWB-based Wald statistic and then sketch its power curve. To conserve

space, here we only present the comparison of power curves of the four methods when $\mu = 0.05$, $c \in \{0, 2\}$, and $\tau \in \{0.05, 0.5, 0.95\}$ which respectively represent the lower, middle, and upper quantile positions.⁴ The two panels in Figure 5, respectively denote the IID and conditional heteroscedasticity

⁴The result for $\mu = 0$ is similar and documented by Figure A1 in the supplementary appendix.

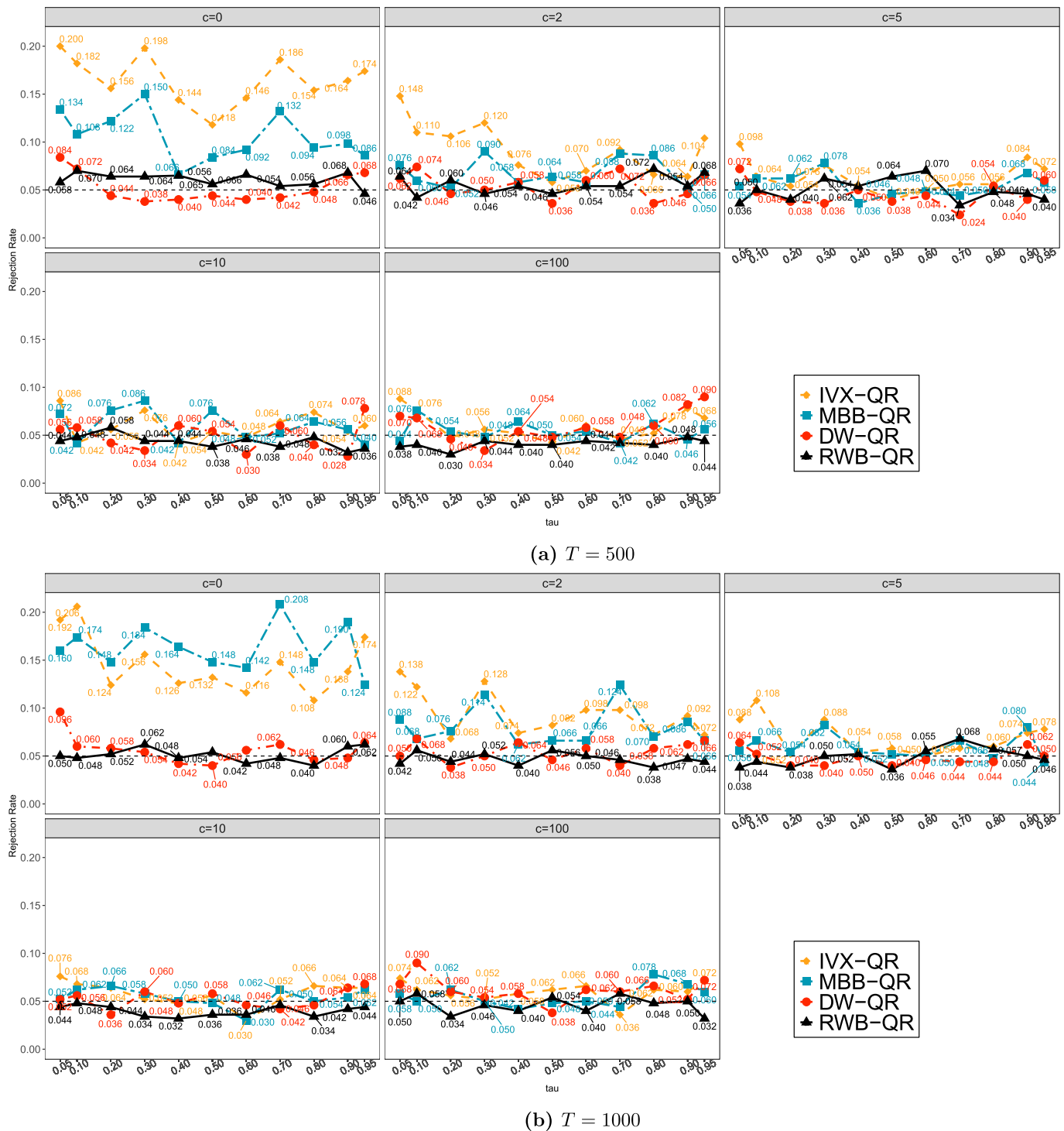
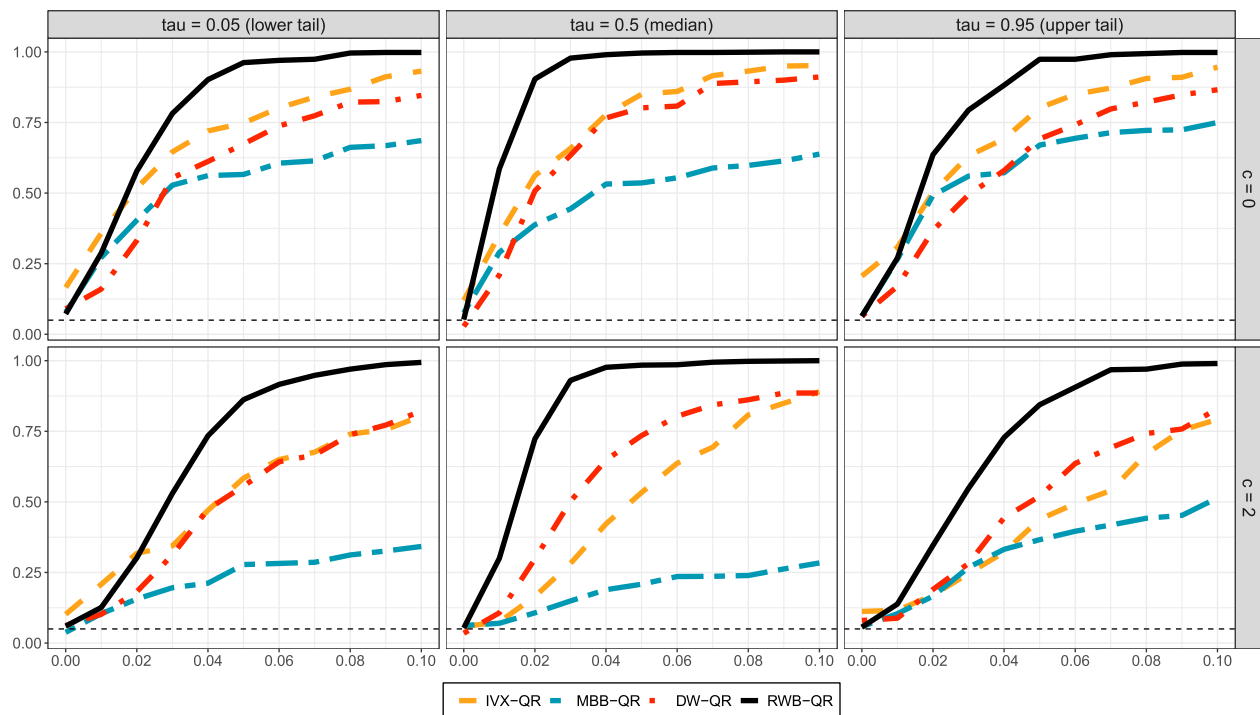


Figure 4. Comparison of size: Conditional heteroscedasticity and nonzero intercept ($\mu = 0.05$).

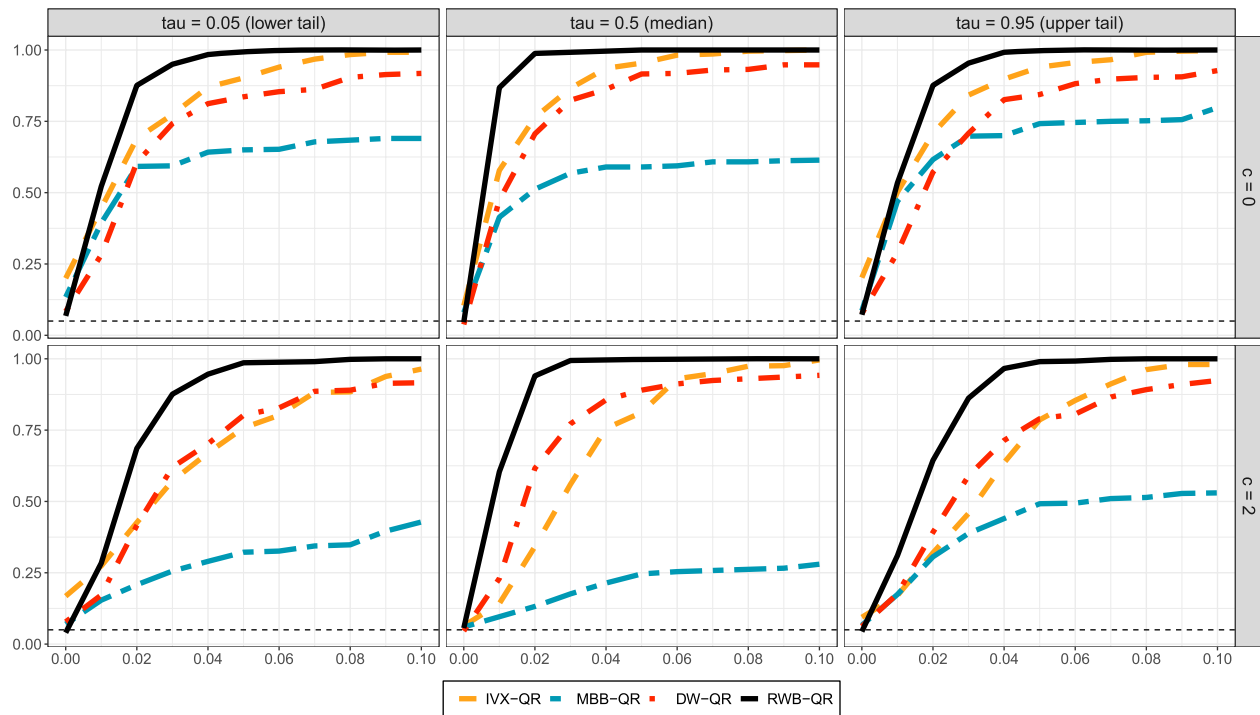
NOTE: This figure displays the size performance of RWB (black), IVX (yellow), MBB (blue), and DW (red) for different τ -quantiles under the conditional heteroscedasticity setting with a nonzero intercept ($\mu = 0.05$) in the predictor. The sample size $T = 500$ (upper panel) and 1000 (lower panel). For RWB, the number of bootstraps is 5000. Each simulation is repeated 1000 times.

scenarios. As seen therein, in both cases, the power by RWB-QR (black) is uniformly larger than that of the three competing methods and grows rapidly as the slope coefficient β_τ increases from 0 to 0.1 at all three quantile positions. IVX-QR and DW-QR also exhibit reasonably good power performance, especially under the conditional heteroscedasticity scenario, but the former displays severe size distortion at both tails when $\beta_\tau = 0$.

In the supplementary appendix, we consider two additional scenarios. First, Cai, Chen, and Liao (2022) further explore the performance of DW-QR when the predictor is near unit root with $c < 0$, and find it works well in this mildly explosive case. Such a scenario is also empirically relevant as many financial and economic variables indeed display exuberant behaviors, as documented by Phillips, Shi, and Yu (2015) and the reference



(a) IID



(b) Conditional heteroscedasticity

Figure 5. Comparison of power: IID and conditional heteroscedasticity with nonzero intercept ($\mu = 0.05$).

NOTE: This figure displays the power performance of RWB (black), IVX (yellow), MBB (blue), and DW (red) for the lower ($\tau = 0.05$), middle ($\tau = 0.5$), and upper ($\tau = 0.95$) quantile positions under the IID (upper panel) and conditional heteroscedasticity (lower panel) scenarios with a nonzero intercept ($\mu = 0.05$) in the predictor and the slope coefficient β ranging between 0 and 0.1. The predictor is unit root ($c = 0$) and nearly integrated ($c = 2$). The sample size $T = 500$. For RWB, the number of bootstraps is 5000. Each simulation is repeated 1000 times.

therein. Following Cai, Chen, and Liao (2022), we investigate the performance of RWB when the predictor is near unit root with $c = -2.5$ when $T = 500$ and $c = -5$ when $T = 1000$ so that $1 - c/T = 1.005$ in both scenarios. Appendix Figure A2

shows that, regardless of the presence of an intercept in the mildly explosive predictor, both RWB-QR and DW-QR display excellent size control, while IVX-QR and MBB-QR become severely oversized. The size distortion exhibited by MBB-QR

Table 1. OLS estimates and unit root tests for eleven predictors.

Predictor	$\hat{\mu}$	$t_{\hat{\mu}}$	$\hat{\phi}$	ADF	DF-GLS	PP	KPSS
Dividend payout ratio (d/e)	−0.0118	−1.729*	0.9848	−3.989***	−3.328***	−3.703***	0.087
Long-term yield (lty)	0.0002	0.510	0.9942	−1.542	0.470	−1.138	0.727***
Dividend yield (d/y)	−0.0368	−2.007**	0.9907	−1.833	−0.049	−1.920	1.481***
Dividend-price ratio (d/p)	−0.0350	−1.896*	0.9912	−1.661	−0.086	−1.825	1.469***
T-bill rate (tbl)	0.0003	0.978	0.9874	−3.177**	−0.318	−2.246	0.448***
Earnings-price ratio (e/p)	−0.0524	−2.434**	0.9830	−3.826***	−0.819	−3.106**	0.795***
Book-to-market value ratio (b/m)	0.0032	1.504	0.9876	−2.140	0.332	−2.625*	1.585***
Default yield spread (dfy)	0.0004	2.674***	0.9636	−3.683***	−2.721***	−3.561***	0.759***
Net equity expansion (ntis)	0.0000	0.158	0.9800	−2.932**	−3.880***	−2.732*	0.483***
Term spread (tms)	0.0013	3.278***	0.9434	−3.723***	−1.556	−3.910***	0.180***
Inflation rate (inf)	0.0011	6.853***	0.5516	−12.697***	−0.085	−11.502***	0.181***

NOTE: This table documents four unit root tests' results for 11 predictive variables sampled from January 1980 to December 2019: dividend payout ratio (d/e), long-term yield (lty), dividend yield (d/y), dividend-price ratio (d/p), T-bill rate (tbl), earnings-price ratio (e/p), book-to-market value (b/m), default yield spread (dfy), net equity expansion (ntis), term spread (tms) and inflation rate (inf). $\hat{\mu}$ and $\hat{\phi}$ are least squares estimates in the AR(1) process: $x_t = \mu + \phi x_{t-1} + e_t$. $t_{\hat{\mu}}$ is the corresponding t statistics for $\hat{\mu}$. ADF represents the statistic of the Augmented Dickey-Fuller test by Said and Dickey (1984) for the null that x_t has a unit root. DF-GLS represents the statistic from an ADF-type test by Elliott, Rothenberg, and Stock (1996) for the null that x_t has a unit root. PP refers to the statistic from the Phillips-Perron test by Phillips and Perron (1988) for the null that x_t has a unit root. KPSS refers to the statistic from Kwiatkowski et al.'s (1992) unit root test for the null that x_t is stationary. The optimal length of lag is determined by the Bayesian information criterion (BIC) for the ADF and DF-GLS tests. *, **, and *** indicate the rejection of the null hypothesis at 10%, 5%, and 1%, respectively.

is even more salient under the conditional heteroscedasticity scenario. For power performance, Appendix Figures A3 and A4 indicate that the power by RWB-QR still dominates the other three competing methods in both scenarios when the predictor is mildly explosive, regardless of the presence of an intercept. Second, we compare RWB-QR with DW-QR by considering a bivariate QR model defined as $y_t = \alpha_\tau + \beta_\tau x_{t-1} + \phi_\tau z_{t-1} + u_{t\tau}$, where $x_t = \mu + (1 - c/T)x_{t-1} + u_{xt}$ and $z_t = \mu + 0.5z_{t-1} + u_{zt}$, so that the persistent level of x_t varies, while z_t is stationary. To conserve space, here we only consider the conditional heteroscedasticity scenario. Specifically, we assume that u_t follows a GARCH(1,1) process such that $u_t = \sigma_t \eta_t$ and $\sigma_t^2 = \omega_0 + \omega_1 u_{t-1}^2 + \omega_2 \sigma_{t-1}^2$, and $\{(\eta_t, u_{xt}, u_{zt})^\top\}$ is a sequence of independent and identically distributed random vectors with a normal distribution with means zero and covariance matrix $\Sigma = \begin{pmatrix} 1 & -0.78 & 0.4 \\ -0.78 & 1 & 0.21 \\ 0.4 & 0.21 & 1 \end{pmatrix}$, which is the same as that of Cai, Chen, and Liao (2022). Except for this, all other simulation settings are the same as those in the univariate case (ii) above. Appendix Figure A5 shows that, regardless of the persistent level of x_t and the presence of an intercept, RWB-QR for multiple in Remark 2 displays excellent size control under the null of $H_0 : \beta_\tau = \phi_\tau = 0$. Furthermore, the power performance demonstrated by Appendix Figure A6 indicates that RWB-QR still dominates DW-QR across all scenarios when both β_τ and ϕ_τ increase from 0 to 0.1.⁵

4. Empirical Example

In this section, we revisit the empirical results of stock return predictability for financial and macroeconomic variables by conducting the proposed RWB-QR method. Similar to the past literature, here we investigate the predictability of 1980–2019 monthly S&P 500 value-weighted log excess returns by examining 11 financial and macroeconomic variables studied by Welch

and Goyal (2008): Dividend payout ratio (d/e), long-term yield (lty), dividend yield (d/y), dividend-price ratio (d/p), T-bill rate (tbl), earnings-price ratio (e/p), book-to-market value (b/m), default yield spread (dfy), net equity expansion (ntis), term spread (tms), and inflation rate (inf).⁶ Detailed definitions of these predictors are available in Welch and Goyal (2008).

We first examine the least squares estimates of the 11 predictors in an autoregressive regression $x_t = \mu + \phi x_{t-1} + e_t$. Table 1 documents the least squares estimates of the intercept and slope coefficient and reveals two observations. First, according to the t -statistics associated with these intercepts, of the 11 predictors, seven (d/e, d/y, d/p, e/p, dfy, tms, and inf) are significant at the conventional levels, suggesting the presence of nonzero intercepts in these series. Second, except for inf, the remaining 10 predictors are highly persistent with the autoregressive coefficient ϕ close to unity. To formally examine the existence of a unit root within these predictors, we apply four widely used tests: the augmented Dickey-Fuller (ADF) test (Said and Dickey 1984), the DF-GLS test (Elliott, Rothenberg, and Stock 1996), the Phillips-Perron (PP) test (Phillips and Perron 1988), and the Kwiatkowski, Phillips, Schmidt and Shin (KPSS) test (Kwiatkowski et al. 1992). ADF, DF-GLS, and PP tests have the null hypothesis that a unit root exists, while the KPSS test has no unit root. For ADF and DF-GLS, the optimal length of the lagged differenced terms is determined by the Bayesian information criterion (BIC). Results in Table 1 suggest that the four tests agree on the existence of a unit root in lty, d/y, and d/p but yield inconsistent results for the remaining predictors.

Table 2 documents some summary statistics of $\hat{u}_{t\tau}$, the estimated residuals of the predictive QR model for the τ th quantile, and $\hat{\eta}_t$, the estimated residuals of GARCH(1,1). For each τ , the first two rows display results of the Ljung-Box test (Ljung and Box 1978) with the null hypothesis of no serial autocorrelation

⁵We thank one anonymous reviewer for suggesting a comparison of the two methods under a bivariate predictive QR model.

⁶The data can be downloaded from a link given in the article titled "A Comprehensive Look at the Empirical Performance of Equity Premium Prediction" at Professor Amit Goyal's homepage: <https://sites.google.com/view/agoyal145>.

Table 2. Summary statistics for estimated $u_{t\tau}$ and η_t .

τ		d/e	lty	d/y	d/p	tbl	e/p	b/m	dfy	ntis	tms	inf
0.05	$LB(\hat{u}_{t\tau})$	9.70	7.57	7.58	7.69	7.50	8.51	7.44	7.75	9.44	7.31	7.87
	$LB(\hat{u}_{t\tau}^2)$	21.29**	7.71	9.19	8.68	7.76	11.00	8.13	15.57	10.71	7.79	8.09
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.99***	-0.07	-0.51***	-0.67***	-0.10**	0.00	-0.01	-0.04
0.1	$LB(\hat{u}_{t\tau})$	11.85	7.29	8.68	7.94	7.53	10.88	7.40	7.79	8.15	7.43	8.06
	$LB(\hat{u}_{t\tau}^2)$	46.40***	9.40	11.94	10.32	10.60	23.59**	9.64	22.43**	10.77	8.89	9.82
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.99***	-0.08*	-0.51***	-0.67***	-0.10**	0.00	-0.01	-0.04
0.2	$LB(\hat{u}_{t\tau})$	7.28	7.88	7.33	7.36	9.26	7.24	7.33	7.01	8.18	7.49	7.17
	$LB(\hat{u}_{t\tau}^2)$	18.43	18.17	13.18	13.19	18.89*	16.75	14.15	19.95*	13.21	12.41	14.70
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.99***	-0.07	-0.52***	-0.67***	-0.10**	0.00	-0.01	-0.04
0.3	$LB(\hat{u}_{t\tau})$	7.22	8.74	7.33	7.36	7.94	7.21	7.38	7.01	7.98	7.32	7.16
	$LB(\hat{u}_{t\tau}^2)$	23.50**	25.43**	18.38	18.68*	23.89**	21.84**	20.32*	25.38**	18.86*	19.37*	23.84**
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.99***	-0.08	-0.52***	-0.67***	-0.10**	0.00	-0.01	-0.04
0.4	$LB(\hat{u}_{t\tau})$	7.30	8.38	7.28	7.30	8.64	7.23	7.86	7.01	7.35	7.32	7.18
	$LB(\hat{u}_{t\tau}^2)$	27.18***	28.55***	26.88***	27.00***	28.83***	27.84***	26.35***	26.57***	27.64***	27.46***	33.32***
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.99***	-0.07	-0.52***	-0.66***	-0.10**	0.00	-0.01	-0.04
0.5	$LB(\hat{u}_{t\tau})$	7.34	7.39	7.28	7.29	7.44	7.27	7.39	7.23	7.36	7.30	7.18
	$LB(\hat{u}_{t\tau}^2)$	29.32***	29.68***	29.37***	29.45***	30.19***	29.48***	28.95***	28.89***	30.17***	29.47***	31.94***
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.12**	-0.03	-0.99***	-0.08*	-0.52***	-0.67***	-0.11**	0.00	-0.01	-0.04
0.6	$LB(\hat{u}_{t\tau})$	7.72	7.31	7.29	7.31	7.32	7.25	7.30	7.36	7.32	7.30	7.18
	$LB(\hat{u}_{t\tau}^2)$	30.70***	29.55***	29.80***	30.08***	29.94***	29.67***	29.72***	30.20***	30.03***	29.45***	35.83***
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.12**	-0.03	-0.99***	-0.08*	-0.52***	-0.67***	-0.11**	0.00	-0.01	-0.04
0.7	$LB(\hat{u}_{t\tau})$	8.36	7.30	7.29	7.31	7.29	7.59	7.30	9.32	7.51	7.30	7.26
	$LB(\hat{u}_{t\tau}^2)$	31.53***	27.90***	28.28***	28.57***	27.80***	28.31***	28.10***	37.29***	30.51***	27.93***	28.39***
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.99***	-0.08*	-0.51***	-0.67***	-0.10**	0.00	-0.01	-0.04
0.8	$LB(\hat{u}_{t\tau})$	9.57	7.37	7.31	7.34	7.34	7.41	7.32	16.98	7.62	7.32	7.22
	$LB(\hat{u}_{t\tau}^2)$	32.07***	25.93**	25.94**	26.42***	25.52**	25.71**	25.84**	49.10***	28.57***	25.99**	26.09**
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.99***	-0.08	-0.51***	-0.67***	-0.10**	0.00	-0.01	-0.04
0.9	$LB(\hat{u}_{t\tau})$	12.25	8.29	7.32	8.53	7.60	7.93	7.39	25.92**	7.58	7.36	7.17
	$LB(\hat{u}_{t\tau}^2)$	32.22***	23.34**	20.93*	23.98**	21.78**	22.66**	21.33**	54.63***	23.04**	21.32**	21.88**
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.99***	-0.07	-0.51***	-0.67***	-0.10**	0.00	-0.01	-0.04
0.95	$LB(\hat{u}_{t\tau})$	9.33	8.06	18.73*	23.63**	7.31	8.61	36.54***	43.00***	8.25	7.44	7.33
	$LB(\hat{u}_{t\tau}^2)$	24.29**	20.23*	27.95***	33.28***	18.25	20.90*	47.24***	70.93***	22.95**	19.30*	18.13
	$\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$	-0.11**	-0.11**	-0.03	-0.97***	-0.08	-0.51***	-0.65***	-0.10**	0.00	-0.01	-0.04

NOTE: This table documents summary statistics for $\hat{u}_{t\tau}$, the least squares residual from the predictive regression at quantile τ , and $\hat{\eta}_t$, the residual of the GARCH(1,1) model. LB denotes the test statistic of a serial correlation test proposed by Ljung and Box (1978) with 12 lags with the null hypothesis of no serial correlation. $\rho(\hat{u}_{t\tau}, \hat{\eta}_t)$ represents sample correlation coefficient between $\hat{u}_{t\tau}$ and $\hat{\eta}_t$. *, **, and *** indicate the rejection of the null hypothesis at 10%, 5%, and 1%, respectively.

within $\hat{u}_{t\tau}$ and $\hat{u}_{t\tau}^2$ based on 12 lags. The results indicate that the residuals of the predictive QR model are not serially correlated in most cases but exhibit heteroscedasticity for most predictors when $\tau > 0.2$. In the third row, we present the Pearson's correlation coefficients between $\hat{u}_{t\tau}$ and $\hat{\eta}_t$, and the results of the correlation test with the null that the correlation coefficient is zero. As can be seen therein, d/e, lty, d/p, e/p, b/m, and dfy exhibit a significant nonzero correlation between $\hat{u}_t$ and $\hat{\eta}_t$ for all τ , indicating the potential endogeneity in the predictive QR.

In summary, most predictors are highly persistent with autoregressive coefficients close to unity, and many are endogenous with heteroscedastic errors during the sample period. All these characteristics are sources of size distortion under the ordinary QR framework and thus call for the adoption of the IVX method. However, as shown by the simulation results, the presence of intercepts in these highly persistent predictors would inflate size distortion by IVX at all quantile positions and raise concerns about MBB due to the potential over-rejection at the tails.

Table 3 documents the inference results of the eleven predictors using IVX-QR, MBB-QR, DW-QR, and RWB-QR for different quantiles. Except for MBB-QR, which reports the 95%

percentile interval, we present the remaining three QR tests' results by reporting their p -values with "*", "**", and "***" respectively, implying the rejection of the null hypothesis of no predictability at the 10%, 5%, and 1% levels. The result roughly echoes the findings in the simulation part and the conclusions of Fan and Lee (2019) and Cai, Chen, and Liao (2022). First, IVX-QR tends to reject the null hypothesis of no predictability more often than the other three quantile methods on both tails. This pattern is particularly salient for d/e, d/y, d/p, e/p, b/m, and dfy. These results should not be surprising because all these highly persistent predictors exhibit strong conditional heteroscedasticity and endogeneity, as demonstrated in Tables 1 and 2. Second, MBB-QR yields fewer significant predictors than the other three methods at conventional levels. In fact, except for d/p and b/m, which are significant at the 5% level when $\tau = 0.95$ and 0.05, none of the remaining predictors is significant at the 5% or better levels. This finding should be expected as it resonates with MBB's power performance in the simulations. Third, DW-QR and RWB-QR exhibit similar performance. However, we observe that DW-QR is more likely to reject the null of no predictability for several lower and upper tail quantiles. For example, DW-QR indicates that d/e is significant at the 5% level

Table 3. Comparison of QR tests: S&P 500 (January 1980–December 2019).

τ	d/e				lty				d/y			
	IVX	MBB	DW	RWB	IVX	MBB	DW	RWB	IVX	MBB	DW	RWB
0.05	0.002***	[−0.040, 0.020]	0.082*	0.370	0.577	[−0.329, 0.324]	0.600	0.820	0.000***	[−0.018, 0.019]	0.338	0.373
0.1	0.005***	[−0.042, 0.021]	0.044**	0.618	0.448	[−0.243, 0.229]	0.760	0.572	0.000***	[−0.014, 0.016]	0.048**	0.341
0.2	0.034**	[−0.034, 0.014]	0.169	0.766	0.141	[−0.298, 0.091]	0.094*	0.936	0.000***	[−0.005, 0.013]	0.178	0.918
0.3	0.375	[−0.025, 0.017]	0.300	0.464	0.004***	[−0.239, 0.068]	0.043**	0.747	0.000***	[−0.002, 0.014]	0.787	0.809
0.4	0.995	[−0.017, 0.018]	0.652	0.679	0.057*	[−0.263, 0.049]	0.152	0.838	0.000***	[−0.002, 0.012]	0.254	0.530
0.5	0.620	[−0.017, 0.015]	0.977	0.331	0.093*	[−0.230, 0.068]	0.456	0.670	0.004***	[−0.003, 0.009]	0.520	0.765
0.6	0.134	[−0.017, 0.015]	0.401	0.330	0.221	[−0.199, 0.093]	0.789	0.483	0.000***	[−0.004, 0.007]	0.365	0.747
0.7	0.064*	[−0.021, 0.015]	0.400	0.174	0.846	[−0.193, 0.094]	0.743	0.990	0.008***	[−0.008, 0.008]	0.726	0.343
0.8	0.001***	[−0.017, 0.022]	0.340	0.716	0.893	[−0.142, 0.169]	0.734	0.993	0.001***	[−0.009, 0.005]	0.871	0.812
0.9	0.122	[−0.028, 0.018]	0.029**	0.815	0.874	[−0.198, 0.364]	0.314	0.998	0.006***	[−0.015, 0.009]	0.021**	0.869
0.95	0.392	[−0.031, 0.018]	0.276	0.907	0.077*	[−0.230, 0.516]	0.983	0.985	0.000***	[−0.021, 0.011]	0.020**	0.230
τ	d/p				tbl				e/p			
	IVX	MBB	DW	RWB	IVX	MBB	DW	RWB	IVX	MBB	DW	RWB
0.05	0.384	[−0.155, 0.028]	0.370	0.372	0.689	[−0.310, 0.294]	0.698	0.998	0.000***	[−0.009, 0.034]	0.827	0.094*
0.1	0.006***	[−0.176, 0.043]	0.785	0.552	0.438	[−0.246, 0.242]	0.577	0.698	0.000***	[−0.018, 0.062]	0.884	0.118
0.2	0.284	[−0.128, 0.054]	0.454	0.821	0.142	[−0.231, 0.091]	0.554	0.417	0.001***	[−0.030, 0.061]	0.569	0.671
0.3	0.626	[−0.089, 0.064]	0.256	0.820	0.053*	[−0.252, 0.117]	0.064*	0.673	0.069*	[−0.025, 0.057]	0.420	0.810
0.4	0.458	[−0.053, 0.058]	0.080*	0.744	0.105	[−0.268, 0.069]	0.044**	0.918	0.306	[−0.017, 0.050]	0.120	0.783
0.5	0.175	[−0.031, 0.039]	0.686	0.767	0.156	[−0.188, 0.099]	0.096*	0.498	0.240	[−0.002, 0.011]	0.681	0.954
0.6	0.788	[−0.034, 0.054]	0.778	0.512	0.311	[−0.216, 0.089]	0.339	0.995	0.949	[−0.014, 0.029]	0.958	0.586
0.7	0.493	[−0.045, 0.082]	0.802	0.217	0.773	[−0.249, 0.092]	0.790	0.999	0.533	[−0.025, 0.033]	0.829	0.980
0.8	0.658	[−0.014, 0.106]	0.159	0.771	0.662	[−0.198, 0.090]	0.828	1.000	0.324	[−0.026, 0.027]	0.872	0.997
0.9	0.240	[−0.008, 0.137]	0.059*	0.674	0.791	[−0.211, 0.346]	0.683	0.636	0.292	[−0.032, 0.030]	0.484	0.470
0.95	0.013**	[0.003, 0.131]**	0.583	0.159	0.999	[−0.275, 0.523]	0.235	1.000	0.800	[−0.023, 0.034]	0.990	0.770
τ	b/m				dfy				ntis			
	IVX	MBB	DW	RWB	IVX	MBB	DW	RWB	IVX	MBB	DW	RWB
0.05	0.039**	[−0.339, −0.010]**	0.155	0.993	0.001***	[−4.314, 0.460]	0.199	0.514	0.136	[−0.356, 0.811]	0.120	0.802
0.1	0.386	[−0.032, 0.024]	0.615	0.967	0.000***	[−3.099, 0.200]*	0.039**	0.779	0.195	[−0.468, 0.721]	0.281	0.962
0.2	0.001***	[−0.252, 0.026]	0.363	0.999	0.006***	[−2.524, 0.433]	0.200	0.995	0.424	[−0.458, 0.319]	0.292	0.948
0.3	0.232	[−0.028, 0.022]	0.575	0.996	0.002***	[−1.889, 0.710]	0.061*	0.995	0.614	[−0.390, 0.210]	0.474	0.035**
0.4	0.483	[−0.149, 0.071]	0.141	0.917	0.001***	[−1.640, 0.398]	0.036**	0.968	0.931	[−0.270, 0.196]	0.616	0.376
0.5	0.263	[−0.031, 0.014]	0.371	0.959	0.565	[−1.535, 0.560]	0.559	0.966	0.828	[−0.148, 0.131]	0.661	0.451
0.6	0.328	[−0.079, 0.091]	0.534	0.994	0.568	[−0.834, 0.801]	0.902	0.999	0.978	[−0.192, 0.163]	0.898	0.968
0.7	0.354	[−0.081, 0.115]	0.908	0.996	0.140	[−0.596, 1.375]	0.125	0.687	0.530	[−0.326, 0.267]	0.422	0.995
0.8	0.324	[−0.028, 0.098]	0.902	0.212	0.015**	[−0.440, 2.658]	0.001***	0.980	0.594	[−0.340, 0.221]	0.608	0.992
0.9	0.305	[−0.020, 0.206]	0.637	0.999	0.000***	[−0.197, 2.670]	0.003***	0.938	0.141	[−0.648, 0.311]	0.843	0.834
0.95	0.402	[−0.018, 0.248]	0.756	0.992	0.000***	[−0.174, 3.465]*	0.000***	0.714	0.347	[−0.645, 0.462]	0.408	0.869
τ	tms				inf							
	IVX	MBB	DW	RWB	IVX	MBB	DW	RWB				
0.05	0.707	[−0.741, 0.773]	0.900	0.853	0.504	[−3.233, 2.535]	0.476	0.904				
0.1	0.563	[−0.834, 0.772]	0.539	0.966	0.103	[−2.534, 2.841]	0.339	0.953				
0.2	0.472	[−0.437, 0.419]	0.168	0.980	0.591	[−1.740, 1.885]	0.308	0.969				
0.3	0.748	[−0.494, 0.263]	0.547	0.899	0.202	[−2.355, 0.727]	0.057*	0.949				
0.4	0.843	[−0.410, 0.223]	0.292	0.963	0.136	[−2.282, 0.572]	0.013**	0.687				
0.5	0.779	[−0.226, 0.140]	0.724	0.928	0.328	[−2.212, 0.252]	0.392	0.843				
0.6	0.777	[−0.297, 0.186]	0.969	0.994	0.338	[−1.770, 0.587]	0.020**	0.189				
0.7	0.978	[−0.324, 0.336]	0.990	0.852	0.807	[−1.943, 0.745]	0.748	0.909				
0.8	0.485	[−0.222, 0.472]	0.825	0.999	0.645	[−1.837, 0.851]	0.517	0.282				
0.9	0.237	[−0.269, 0.774]	0.686	1.000	0.441	[−2.644, 1.454]	0.723	0.344				
0.95	0.094*	[−0.235, 0.950]	0.345	0.936	0.999	[−2.138, 2.760]	0.507	0.993				

NOTE: This table compares the predictive power of the eleven predictors based on IVX-QR, MBB-QR, DW-QR, and RWB-QR at different quantile positions. For MBB-QR, the 95% percentile intervals of the estimates are presented. The values denote the p -values rounded to the third decimal place for exposition for the remaining three methods.

*, **, and *** indicate the rejection of the null hypothesis of no predictability at 10%, 5%, and 1%, respectively.

on both tails, and dfy is significant at the 1% level on the upper tail. At the same time, none is significant at any conventional levels by RWB-QR. These conflicting results are in line with the findings in the simulation part, which show that DW-QR tends to be oversized at tails for highly persistent predictors with conditional heteroscedasticity. In sum, the proposed RWB-QR procedure provides a reliable and conservative inferential tool for empirical researchers to test the predictability of financial asset returns.

5. Conclusion

This article develops a new unified inferential theory for predictive quantile regression regardless of zero or nonzero intercept in modeling predictors, while the existing methods require zero intercepts. The test statistic is constructed using a weighted quantile regression estimation, resulting in a normal limit after random normalization. Using the square of a t -statistic with variance estimated by a random weighted bootstrap procedure, we obtain a unified predictability test whose statistic follows

a Chi-squared distribution under the null of no predictability. Compared with other competing approaches, our method offers a robust inference regardless of the presence of intercept and persistence of a predictor. Monte Carlo simulation demonstrates that our method exhibits significantly better size and power performance than the competing methods in different scenarios, particularly when the predictor is highly persistent and contains a nonzero intercept. In the empirical application, our method shows that only net equity expansion (ntis) exhibits a significant predictive capability when $\tau = 0.3$, suggesting that the proposed method is a conservative inferential tool for empirical research in predictive quantile regression.

Supplementary Materials

The supplementary appendix provides the technical proofs of [Theorems 1–2](#) in the main paper and some additional Monte Carlo simulation results.

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